Machine Translation Based on the Relations between Words

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Abstract

This paper describes a new approach in the Example-Based Machine Translation. During the search of the best example matching the input sentence, the relation between two consecutive words is considered as unit of comparison. It strengthens correspondences between words and can find a set of word matches equally divided along the whole sentence. The transfer process is a simple delete, replace and insert operations applied to the selected example. The position of the segment to be inserted will be predicted according to its neighbors in other selected examples. Treating a sentence as series of words is considered to be able overcome imperfectness of the tree dependency structure, on which other methods are based. The system is implemented in French-Japanese machine translation, and spoken language texts are taken as examples.

1. Introduction

To overcome the imperfectness of the rule-based machine translation, we have started a research on the Example-Based Machine Translation (EBMT). EBMT consists of two process: the matching, which searches the examples matching the input sentence, among a set of translation examples, and the transfer, which produces the translation result from the selected examples. The idea was first proposed by Nagao [Nagao 84]. Various models have been proposed [Sato and Nagao 90], [Kitano 93],[Watanabe 95] and different issues are discussed [Jones 96]. As far as word-based matching method are concerned, there are the DP-Matching method, and the "length first" method. During the selection of the best match, using a single word as unit of translation in the DP-Matching [Kitano 93], does not give a set of word matches equally divided along the whole sentence, and can produce mismatch with isolated word match. As for the DP-Matching method proposed by Cranias [Cranias 94], the definition and search of the functional words and phrases do not always succeed. On the other hand, the length first method [Niremburg 93] does not give a way to decide which segment will be chosen first when multiple segments having a same length are equivalents. That case always appears when only part-of-speech, for example, is considered. We have presented a matching method [Andriamanankasina 97], which considers the relation between two consecutive words, as unit of comparison. It strengthens any match found between words and can produce a set of word matches equally divided along the whole sentence. The comparison is based on the part-of-speech which is assumed to be the element which determines the surface syntactic structure of the sentence. As for the transfer method, since the translation of a contiguous segment is not necessarily a contiguous segment in the target sentence, dividing sentence into segments is considered not to be a good method. Definition of the segment will be also an issue to be tackled. Other methods accept a tree dependency structure of words in the sentence as input [Sato 90, Watanabe 95]. However, to overcome fails in the construction of these structures, we propose a method based on a simple delete, insert or replace operation applied to the best matching sentence. The idea inside this method is to select multiple best outcomes during the matching process, and predict the segment position by looking at its neighbors in other examples. The translation system is implemented in French-Japanese machine translation. In the next section, the overview of the whole translation is presented. It consists of three phases: the tokenization and tagging of the input sentence, the matching process and the transfer process, which are described in the following sections. Detail of the experiments is presented and few words are given as conclusion in the last section.

2. Overview

2.1. Overview of the translation

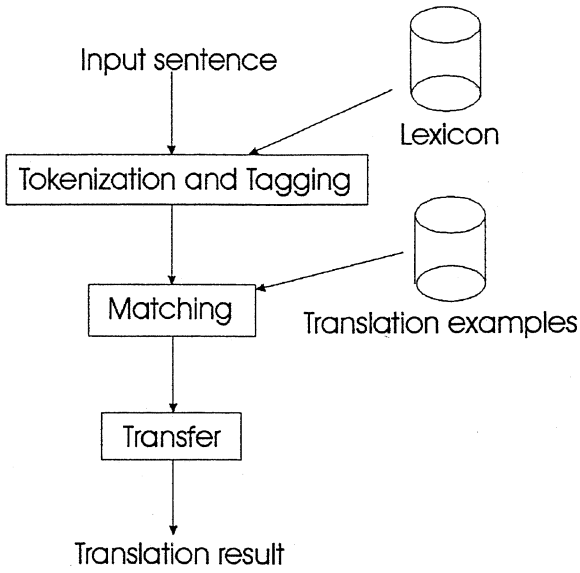


Fig. 1: Overview of the translation

The translation is divided into 3 steps [Fig. 1].

First, the input sentence is tokenized. The goal of the tokenization is to combine words whose set corresponds to one part-of-speech (POS) tag, or separate word in the other case. This is important because sentences are usually written differently. The list of tokens and their POS tags are determined by the lexicon. Disambiguation is not performed at this step and all possible POS tags are attached to each token.

Second, the matching operation is performed. Ten best sentences matching the input sentence are selected from the translation examples. During the same time that the matching operation is performed, the program collects translation examples whose source sentence contains words of input sentences. These sentences are used to translate words or segment of words of the input sentence.

The last step is the transfer operation, which transforms the best match into the target sentence.

2.2. Structure of the translation examples

An entry in the translation examples is presented in Table 1. It is composed by the French source sentence, its Japanese translation sentence, and a map describing the correspondence between words in both sentences. Every word in the French source sentence is followed by its POS tag. As for the Japanese, the structure *"stem pronunciation lemma tags"* from the result of the Japanese tagging program Chasen 1.0 [NARA-AIST 97] is used, for simplicity. The correspondence map is of the form *WPF1,WPF2,.../WPJ1,WPJ2,...*, where *WPFi* is word position in the French sentence, and *WPJj* is word positions in the Japanese sentence. In the example of the Table 1, *2/4,5* means that the word *"sont"* corresponds to あります, *"au"* corresponds to に, and *"premier etage"* to 二階. Because of the non-availability of a good word-alignment program, constructing of the correspondence map is almost done manually. Word or segment having no correspondent is not specified in the correspondence map. Only segments which are formed by contiguous words are considered.

Table 1 : A translation example

French sentence : *elles/PRV sont/ECJ au/DTC premier/ADJ etage/SBC ./.*
Japanese sentence : 二 に 二 数詞 階 かい 階 普通名詞 に に に 格助詞 あり あり あり 動詞　子音動詞ラ行　基本連用形 ます ます ます 動詞性接尾辞 動詞性接尾辞ます形 基本形 。 。 。 句点 EOS
Correspondence maps : *2/4,5 3/3 4,5/1,2*

3. Algorithms

3.1. Tokenization and Tagging

To tokenize the input sentence, the program works as follows:

- Blanks are first put between any punctuation, or a hyphen, and after an apostrophe.
- Then, exceptions are assembled again according to the list of tokens given inside the lexicon.

For example, the sentence:

je viens tout de suite. (私はすぐ来ます)

becomes:

je / viens / tout_de_suite / .

after tokenization.

As for the tagging, disambiguation is not performed at this step and all possible POS tags are collected. For example, the word *"le"* may be a pronoun or a determinant, therefore, from the sentence:

je voudrais appeler le Japon . (私は日本に電話をしたいのです)

the output will be:

je/PRV voudrais/VCJ appeler/VNCFF le/DTN,PRV Japon/SBP ./.

We use a lexicon which is developed by the INALF (Institut National de Langue Francaise) [INALF 97]. It is presented in the Table 2. An entry is composed by the token (word), and the list of its possible POS tags.. INALF proposed 70 POS tags in which 20 are punctuations and 3 are for particular cases. Differences between plural and singular are removed, because they make a negative contribution to capture similarity of two sentences. We use therefore 48 POS tags. They are characterized as follows :

1. Verbs *être* (to be) and *avoir* (to have) are separated from the rest of the verbs. This is justified by the special use of these verbs in the language. Any verb may take 4 tags depending on its form : infinitive, conjugated, past participle, or *-ant* (-ing) form.
2. Combined determinant, like *des, du,* and simple determinant are treated as belonging to different groups.
3. Pronouns are divided in two groups : those supported by the verb, and the others.
4. Simple adjectives are separated from past participle form adjectives, which may take different POS tags depending on whether they follow the verb *être* or not.
5. Among the rest of the tags, there are nouns, adverb, cardinal, preposition, relative.

Table 2: Structure of the lexicon

Word	**POS tags list**
tous	PRO DTN
tout	PRO DTN ADV
tout_au_plus	ADV
tout_de_suite	ADV

3.2. Matching algorithm

The matching consists of two procedures - one for selecting sentences matching the input sentence from the translation examples, and the other for selecting partitions in order to translate segments of the input sentence. These procedures are performed simultaneously.

For the search of the best sentence matching the input sentence or "the best match", the main idea is to find a set of word matches which are equally divided along the whole sentence. A word match represents an equivalence between a word of the input sentence and a word of the example sentence. Any word match must be supported by a match between their direct neighbors. The shortest sentence having the largest number of word matches is chosen as the best match. Word comparison is based on POS tags.

The algorithm can be summarized as follows:

Each word from the input sentence is compared, in turn, with words of the translation example, as in Fig.2. Only equivalence supported by an equivalence between one of the words' direct neighbor is considered. In other words, to decide if a word at a position *i* in the sentence *S1* match a word at the position j in the sentence *S2*, they must be equivalent, and the word at the position *i-1* (or *i+1*) in *S1* must be equivalent the word at the position *j-1* (or *j+1*) of *S2*. Since disambiguation is not performed during the tagging operation, a word *W1* of the input sentence is considered to be equivalent to a word *W2* of the source sentence, if the POS tag of *W2* appears in the POS tags list of *W1*. The result of this operation is a list of word correspondences between the source sentence and the example. It is ordered by *WPI* + *WPS*, where *WPI* is the word position in input sentence and *WPS* the word position in the example.

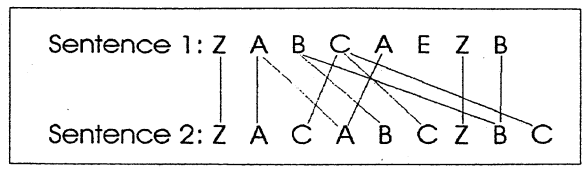


Fig. 2: Comparison of two sentences

To find a non-interfering set of word matches, deleting operation is applied to the word correspondences list, which is represented by the connection lines in Fig 3. "Doubles" or "crossings" are eliminated. Two attempts are considered: deletion starting from the left and one starting from the right. The one which leaves larger number of word matches is chosen. "Double" appears when one word has many correspondences in the other sentence. "Crossing" appears if *WPI1* < *WPI2* but *WPS1* > *WPS2*, where word *I1* has a correspondence *S1*, and word *I2*, *S2*.

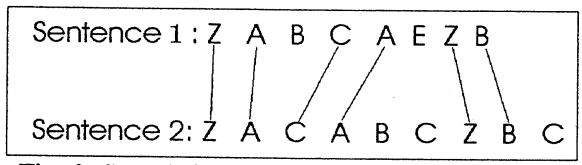


Fig. 3: Search for a non-interfering set of word matches

To find the shortest sentence which have the largest number of word matches, the following metric is proposed:

$$M = 100*NT - 10*L + NW, \quad (1)$$

where, *NT* is the number of word matches found, *L* is the length of the sentence, and *NW* is the number of exact matches. The bigger this value is, the closer the two sentences are. An illustration is presented in Table 3.

Table 3: Calculation of the Similarity Metric

Input: *est/ECJ -ce/PRV que/SUB,SUB$,PRO je/PRV pourrais/VCJ trouver/VNCFF des/DTC cotelettes/SBC de/PREP porc/SBC ?/?*	
Example: *quand/SUB est/ECJ -ce/PRV que/SUB$ vous/PRV passerez/VCJ le/DTN concours/SBC d'/PREP admission/SBC ?/?*	11 words
Word matches found : *est -ce/est -ce, ce que/ce que, que je/que vous, vous pourrais/vous passerez, cotelettes de/concours d', de porc/d' admission, porc ?/admission ?,*	7 word matches
Exact Matches : *est, -ce, que, ?.*	4 matches
Similarity Metric *M = 7*100 - 11*10 + 4 = 594*	

As for the search of the partitions, its main goal is to translate segment of input sentence. Our method selects, per word of the input sentence, a sentence having the longest segment matching the segment starting from that word. Comparison is based on exact match. If exact match expires, POS tags will make the difference.

3.3. Transfer algorithm

The transfer consists of 5 steps. The first two steps are preparations of the sentences. These 5 steps are:

- Selection of the right POS tag for multiple-tagged word
- Segmentation of sentences according to some characteristics of the languages
- Constructing the target sentence by the segments covered in the best match.
- Insertion of segments having no correspondent in the best match
- Translation of each segment

3.3.1. Selection of the definitive POS tag for multiple-tagged word

Ten sentences having the ten highest metrics are selected during the matching. For multiple-tagged word, the POS tag of its correspondent is taken as its definitive POS tag, if it has a correspondent in the best match. As for the example in Table 3, since *que/SUB,SUB$,PRO* corresponds to *que/SUB$*, *SUB$* is chosen as POS tag of *que*. If any correspondent has not been found in the best matching sentence, the next sentence will be used. If these ten selected sentences expires, and any correspondent still has not been found, sentences selected during the search of the partitions are used.

3.3.2. Sentence segmentation

The goal of the segmentation is to be able to delete, or insert word which usually have no correspondent in other language, but tightly closed to other words. To avoid error, only obvious cases are considered, these are particles, determinants, prepositions, prefixes and suffixes. For example, as far as French is concerned, determinants are considered to be with the word following them, therefore, the phrase:

la semaine prochaine . (来週)

becomes:

/la semaine/ prochaine/ ./

after segmentation.

As for the Japanese language, the sentence:

日本 の 映画

becomes:

/日本 の/ 映画/

Having the sentences segmented, correspondences between words in two sentences (input and example, or source and target) can be transformed into correspondences between segments.

3.3.3. Constructing the target sentence by segments which are covered in the best match

The translation result is sketched by the segments of input sentence which are covered by the best match. The translation of the best matching sentence is used as reference. Segments having no correspondence in the input sentence are deleted, as in Fig. 4. If there is an exact match between a segment of the input sentence and a segment of the best matching sentence, its translation in the target sentence is kept.

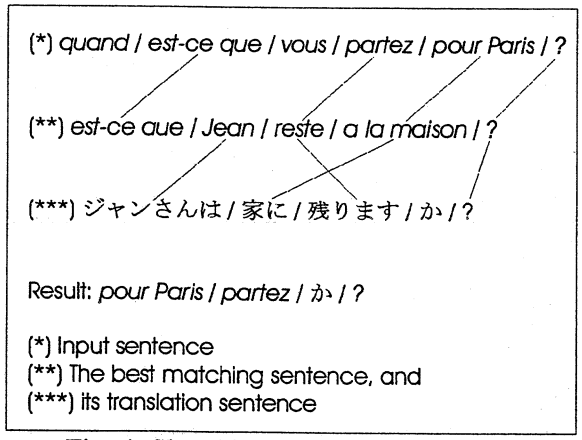


Fig. 4; Sketching the translation result

3.3.4. Insertion of segments uncovered by the best matching sentence

For segments which do not have correspondent in the best matching sentence, other matching sentences are used. We assume that they have correspondents in other matching sentences. The insertion starts with the second best matching sentence and continues until the tenth. To find where the segment should be placed, the closest segment having a correspondent in the input sentence is selected, and the segment to be inserted is positioned before or after that segment. The process is described in Fig. 5. どこへ precede 今晩 in the Japanese sentence, therefore *"quand"* is put after *"pour Paris"*.

3.3.5. Segment translation

Having the examples selected during the search of partitions, the segment meaning can be obtained easily: One sentence per word of input sentence is selected. For example, if the sentence *"quand sera la prochaine fete ?"* (次の祭りはいつですか?) is selected for the first word *"quand"*, the meaning of the segment *"quand"* (いつ) can be deduced by segmentation since the mapping between words of the source sentence and target sentence is given within the example translation. For segment composed by 2 or multiple words, all fixed words in the selected partition must match exactly with the segment of the input sentence. We assume that the selected partition gives a segment matching the segment of the input sentence. As for the example of Fig. 5, after this step, the system will find the translation of each segment : *quand*/いつ, *vous*/あなたは, *partez*/行きます, *pour Paris*/パリへ. Therefore, the translation result will be: あなたはパリへいつ行きますか?

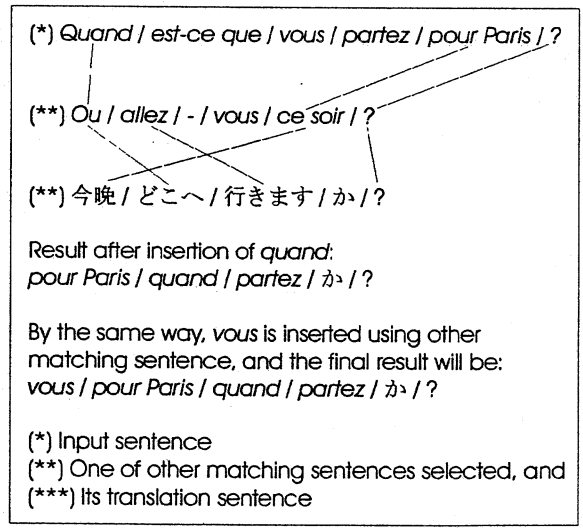


Fig. 5: Insertion of other segment

4. Experiments

4.1 Methods

The translation examples contains 1098 French-Japanese pairs, taken from French-Japanese

Conversation books [Meguro 87] and [Sato F. 90]. One French sentence has an average of 8 words. It is justified by the presence of a short sentence in the spoken-language world, like *Quoi ?* (What ?). The longest sentence has 19 words. We train the French source sentences of the translation examples, with the tagging program developed by the INALF, and the Japanese sentence with the Japanese language tagging program [NARA-AIST 97]. The sub-sentential alignments was done manually, since a very large corpora is necessary for automatic alignment [Gale 91, Fung 94, Brown 93]. The system is tested with 367 French input sentences, taken arbitrarily from the same source. We have observed the rate of the translation, the success rate of segment positioning during the transfer, and the reason of the fails sentence per sentence. To evaluate the translation result, since unregistered words exist, we focus our observation especially on segment position and consider the translation as "correct" if all segments are put at their right places.

4.2. Results

The result of the experiments is presented in Table 4. It shows that the method gave 52.0% of correct translation.

Table. 4: Result of the translation

Correct Translation	191 sentences (52.0 %)
Failed Translation	176 sentences (48.0 %)

The effectiveness of the segment insertion using other matching sentences is presented in Table 5. Among the correct translation result, 51.3% are sentences which were not covered by the best matching sentence itself. In other word, 51.3% of the correct translation are saved by the segment insertion method proposed for the transfer.

Table 6: Effectiveness of the segment insertion using other sentences during the transfer

Presence of segment insertion	98 sentences (51.3%)
No segment insertion	93 sentences (48.7 %)
Total	191 sentences

Some reasons of the fails are presented in the Table 6. "Others" in the last row represent especially fail of the segment translation, and presence of unnecessary words in the translation result. Fails of segment translation are closely related to word meanings. For example, to translate the segment "*il y a*" in the phrase *"il y a 20 kilometres"*(２０キロメートルあります), if the phrase *"il y a une semaine"* (一週間前に) is selected, the translation result will be２０キロメートル前に. As for the presence of unnecessary words, these are words in the Japanese sentence, which do not have correspondent in its French translation, as the case of the segment か in the following case:

Input sentence:
"je vais au cinema" (私は映画館に行きます),
Best matching sentence:
"quand est-ce-que vous allez au cinema ?" (いつあなたは映画館に行きますか?)
Translation result:
私は映画館に行きますか
However the right translation result is:
私は映画館に行きますか.
Although these words helps in some cases, they are not necessary in others.
The "Presence of segment not covered" in the third row is the case of sentence whose all segments could not be covered by the 10 matching sentences selected. Uncovered segment does not appear in the translation result.

Table 6: Reasons of the fails

Fail because of the quality of the best matching sentence selected	46 sentences (26.1 %)
Fail only by segment insertion	20 sentences (11.4 %)
Presence of segment not covered	24 sentences (13.6 %)
Others	86 sentences (48.9 %)
Total	176 sentences

4.1 Discussion

As far as the number of the translation examples is considered, 52.0% of a correct translation rate can be considered as very promising result. From the fact that only 11.4% of the fails were from the fails of the segment insertion method itself, we

can also confirm the effectiveness of this method for the transfer.

Compared to earlier experimental research on EBMT [Sato 90][Kitano 93][Watanabe 95], we , in this research, consider a sentence as a finite series of words and avoid dealing with the tree structure of the sentence. In addition, we have proved in our previous work [Andriamanankasina 97] that considering 2 consecutive words as unit of comparison during the matching, gives better result that comparison based on single word. That is always true as far as the surface syntactic structure of a sentence, not partition, is searched.

The very obvious improvement of this method is of course the multiplication of the translation examples. However, there are few points which need to be commented here:

First, the POS tags list. The larger the number of POS tags is, the better the result of the matching is. However, that is true only in the case where the desired best matching sentence is present in the translation examples. We have for example 20 punctuations and symbols, separation of single determinant and compound determinant, to name a few. Before making any conclusion, we are planning to do multiple experiments with different composition of the POS tags list.

Second, the tagging of the input sentence. The tagging method which is used in our method can be called example-based tagger. It can be replaced by any available tagger which gives nearly perfect result. Only the processing time makes the difference.

Third, the fails of the segment insertion. We assume that contiguous segments are related between themselves. In the example of Fig.5, 行きます which is the segment following どこへ, does not have a correspondent in the input sentence, therefore the relation between どこへ and 今晩 is only considered. However, including the case where any contiguous segments are not found, different possibilities might appear. The success of the translation depends on the decision which must be taken for each situation. In addition, to search the best place for any uncovered segment, using partial matches, or partitions, seems to be more appropriate. The problem of uncovered segment will also be cleared, since in that case it is probable to find multiple partitions which can cover the input sentence. However, as the case of *"Ou"* and *"ce soir"* in the example of Fig. 5, although their correspondents are connected in the target sentence (どこへ and 今晩), they are separated far away in the source sentence. The chance to find a partial match which can cover long segment will be very low, if multiple shorter matches are preferred over long segment match.

Fourth, our research is limited in the contiguous segment. To align the phrase *"ne va pas"* with 行きません, the obvious way might be aligning *"ne pas"* with ません and *"va"* with 行き [Brown 93]. We assume that there is always a way to modify any non-contiguous segment map into a contiguous one by combining segments. Only one map between *"ne va pas"* and 行きません is therefore considered. Although *"va"* alone cannot be translated, a contiguous segment map is obtained.

Before giving any comment on the segment translation, we prefer doing further experiments, since the number of translation examples is very limited and dictionary is not used. For segment composed by multiple words, the algorithm can still be improved. Since only short segment is considered, we assume that a good match can always be found. Segment combination can also be conceivable if longer match is found and the segments are still related after sketching.

5. Conclusion

We have presented in this paper a new approach in the example-based machine translation. During the search of the best sentence matching the input sentence, it considers the relation between two consecutive words as unit of comparison. In addition, the sentence is viewed as a finite series of words and any attempts to find the tree structure of the sentence is avoided. The transfer is an operation of deletion, modification and insertion of segments, which are applied to the best matching sentence. Uncovered segments are inserted according to segments relation which may be defined in other matching sentence. Result of the experiments are very promising, as far as the number of the translation examples are considered. To improve the results, a study with larger data, improvement of the POS tags list, the segment insertion and translation are necessary.

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